

# SHARANYA MISHRA

Chicago, IL | (312) 292-6143 | [sharanyamishra8@gmail.com](mailto:sharanyamishra8@gmail.com) | [LinkedIn- Sharanya Mishra](#) | [GITHUB – mishrasharanya](#)  
Portfolio: <https://www.sharanya-mishra.com/>

## EDUCATION

Illinois Institute of Technology, Chicago, IL May 2026  
**Master of Science- Data Science - GPA: 3.7**  
PES University, Bengaluru, India June 2024  
**Bachelor in technology - Computer Science/Engineering-GPA: 3.5**

## SKILLS

**Programming & Tools:** Python, R, SQL, C++, JavaScript, MATLAB, Git, Docker, Kubernetes  
**AI / ML & Data Science:** Machine Learning, Deep Learning, NLP, Computer Vision, Regression, Time-Series Forecasting (ARIMA, SARIMA, GARCH), Feature Engineering, Model Evaluation  
**Data Analysis & Statistics:** Probability, Hypothesis Testing, Statistical Inference, Process Optimization, Work Measurement, A/B Testing  
**Visualization & Reporting:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Excel (Pivot Tables, VLOOKUP, Variance Analysis), SAP  
**Cloud & Big Data:** AWS (S3, EC2, SageMaker), Microsoft Azure, PySpark, Hadoop, Jenkins

## EXPERIENCE

**Lead Research Assistant** January 2025 - Present  
*Illinois Institute of Technology*

- Built a **React + Node.js trajectory optimization platform** translating robotics problems into **solver-ready** formulations across 50+ NEOS solvers
- Improved **simulation performance by ~30%** through real-time computation, optimized workflows, and modular execution logic
- Integrated **LLM-based** modules to support adaptive optimization setup, interpretation, and user-driven problem configuration

**Data Science Consultant / Project Lead** January 2026 – Present  
*LabelMaster Practicum*

- Analyzed multi-year customer, transaction, product, and location datasets (2012–2026) to study **churn, retention, and revenue concentration**
- Identified a key trend: active customers declined while revenue per customer increased, highlighting **account concentration risk**
- Built **customer segmentation tiers** using revenue contribution, purchase activity, geography, and dormant-account behavior
- Found win-back opportunities among inactive but historically valuable customers across priority states.
- Presented stakeholder-ready recommendations for retention campaigns, geographic targeting, and customer reactivation strategy

**Data Classification- Spring Intern** January 2024 - July 2024  
*Aptos Retail*

- Built **15+ Azure-based data validation** rules to classify sensitive enterprise data and improve reporting accuracy to **95%**
- Automated classification workflows across **5+ teams** using Microsoft Purview and Salesforce, reducing manual effort by **40%**
- Supported **data governance** by improving data visibility, ownership tracking, and compliance-ready reporting

**Data Analyst Summer Intern (Risk Advisory)** June 2023 - August 2023  
*Deloitte Touche Tohmatsu LLC*

- Automated** anomaly detection across **7+ client datasets** using **Python** and **SQL** to identify reporting inconsistencies
- Built Excel dashboards using **Pivot Tables, VLOOKUP**, and variance analysis to support risk advisory decisions
- Cleaned and validated client data to improve audit-readiness and reporting efficiency

**Data Analyst Intern** June 2022 - August 2022  
*Capgemini*

- Developed **SQL-based analytics** pipelines and **Tableau dashboards**, enhancing client reporting efficiency.

## RESEARCH PUBLICATION

**Audio Description for the Visually Impaired (IEEE, 2024)**  
**Link to paper:** <https://ieeexplore.ieee.org/document/10544216>  
**GitHub:** [https://github.com/Obsarian/Audio-Description-of-Videos\\_Capstone](https://github.com/Obsarian/Audio-Description-of-Videos_Capstone)

- Developed a **real-time video captioning pipeline** using YOLO, ResNet-35, and Google Text-to-Speech API
- Combined **object detection + action recognition** to generate more accurate scene-level audio descriptions
- Achieved BLEU score of 0.7 after evaluating **100+ video samples** for caption quality

## PROJECT

**HerWay -SoReMo Fellowship**  
**GitHub:** <https://github.com/mishrasharanya/HerWay-Soremo>

- Built a unique **multi-source urban analytics system** combining Reddit, 311, and crime datasets to generate neighborhood-level insights across **77 Chicago community areas**
- Developed one of the only pipelines in the fellowship integrating NLP and the **OpenAI API** to extract community signals from **1,000+ public records/posts**
- Created a **chatbot + interactive map** to convert raw data into **actionable indicators** for safety perception, service issues, and neighborhood-level decisions

**ForeQuest – Volatility Forecasting and Option Pricing App**  
**GitHub:** [https://github.com/mishrasharanya/Forequest-Forecast\\_Smarter](https://github.com/mishrasharanya/Forequest-Forecast_Smarter)

- Built an end-to-end **financial forecasting app** using **ARIMA, GARCH, Monte Carlo simulation, and yFinance** to predict stock trends and volatility
- Achieved **~85% R<sup>2</sup>** and **<0.02 MAE** in volatility prediction with model evaluation across MSE, RMSE, MAE, and directional accuracy
- Integrated **LLM-based summaries** to explain forecast outputs, volatility shifts, and **option-pricing results** in plain language for users

**InsightFlow – Data Pipeline & Analytics System**  
**GitHub:** <https://github.com/mishrasharanya/insightflow-pipeline>

- Built an end-to-end **ETL pipeline** to ingest, clean, validate, and transform multi-source data into **analytics-ready tables**
- Designed dashboard views for real-time trend monitoring, data quality checks, and business insight generation